

Table 1: Causal hyperparameter-tuning battery: does causally optimized HP selection beat a fixed-HP incumbent? Every arm re-selects its hyperparameters *periodically and causally*: every 250 solves, a leakage-clean forward split of the current 500-trading-day (24,000-bar) training window — fit block, 25-bar embargo (covering the 25-bar HAR moving average), 125-bar validation tail — re-selects the configuration by validation MSE, so no forecast’s hyperparameters ever see its future. Linear arms (ridge / recursive lasso / recursive elastic net) search log-scale penalty grids and slide warm between tunings (exact rank-1 / online-homotopy updates, cold reseed at each tune); tree arms (LightGBM, XGBoost) search max-depth $\{3, 5, 8\} \times \text{colsample} \{0.25, 0.5\}$ with learning rate pinned at the production 0.1, grid cells fit at 300 trees, the winner refit every bar at the production 500 trees. Model refit is every bar throughout. Full out-of-sample panel ($n = 218,934$ thirty-minute bars; identical rows for every cell), computed as 100 walk-forward chunks with a 24,000-bar warm-up halo and pooled by concatenation. The incumbent is OLS on HAR + calendar only (no exogenous features; the production ridge path at $\alpha = 0$, exact OLS, per-bar refit), one shared comparator for every arm. DM = Diebold–Mariano test on per-bar QLIKE loss vs. that incumbent, reported as (t, p) with HAC standard errors (automatic lag); negative t favors the tuned arm. QLIKE, evaluated on the raw-variance scale, is the primary level-robust ranker; raw-variance MSE and OOS- R^2 are dominated by a small number of COVID bars. Unlike earlier decks, OOS- R^2 here is measured against the shared incumbent (not the unconditional mean), so the incumbent’s own cell is undefined. Best QLIKE and best OOS- R^2 in bold. All 45 arms (5 estimators $\times 9$ buckets) are complete.

exogenous bucket	estimator	QLIKE	MAE	MSE	OOS- R^2	MZ- β	MZ- R^2	DM vs. incumbent (t, p)
all_features	LightGBM causal-tuned	0.12603	1.222×10^{-6}	4.080×10^{-11}	0.0590	0.891	0.7230	$-15.4, 1.1 \times 10^{-53}$
all_features	XGBoost causal-tuned	0.12640	1.231×10^{-6}	4.605×10^{-11}	-0.0621	0.840	0.7007	$-16.5, 1.7 \times 10^{-61}$
all_features	ridge causal-tuned	0.12787	1.242×10^{-6}	4.586×10^{-11}	-0.0575	0.884	0.6887	$-19.9, 2.4 \times 10^{-88}$
moments	LightGBM causal-tuned	0.12815	1.230×10^{-6}	4.245×10^{-11}	0.0209	0.882	0.7133	$-11.2, 1.9 \times 10^{-29}$
all_features	ElasticNet causal-tuned	0.12827	1.245×10^{-6}	4.260×10^{-11}	0.0174	0.904	0.7076	$-19.3, 4.1 \times 10^{-83}$
liquidity	LightGBM causal-tuned	0.12839	1.236×10^{-6}	4.086×10^{-11}	0.0576	0.892	0.7226	$-14.6, 7.1 \times 10^{-49}$
moments	XGBoost causal-tuned	0.12862	1.235×10^{-6}	4.151×10^{-11}	0.0425	0.874	0.7221	$-9.0, 1.7 \times 10^{-19}$
liquidity	XGBoost causal-tuned	0.12886	1.240×10^{-6}	3.966×10^{-11}	0.0853	0.879	0.7346	$-13.9, 2.2 \times 10^{-44}$
all_features	Lasso causal-tuned	0.12906	1.244×10^{-6}	4.232×10^{-11}	0.0239	0.915	0.7078	$-13.2, 5.9 \times 10^{-40}$
market_vw	LightGBM causal-tuned	0.12963	1.247×10^{-6}	4.378×10^{-11}	-0.0096	0.861	0.7098	$-10.8, 2.1 \times 10^{-27}$
vol_demand	LightGBM causal-tuned	0.12979	1.253×10^{-6}	4.486×10^{-11}	-0.0346	0.860	0.7023	$-10.0, 1.5 \times 10^{-23}$
implied_vol	LightGBM causal-tuned	0.13000	1.262×10^{-6}	4.483×10^{-11}	-0.0338	0.840	0.7099	$-7.8, 4.5 \times 10^{-15}$
market_vw	XGBoost causal-tuned	0.13011	1.251×10^{-6}	4.495×10^{-11}	-0.0367	0.854	0.7034	$-8.0, 7.1 \times 10^{-16}$
vol_demand	XGBoost causal-tuned	0.13026	1.266×10^{-6}	4.810×10^{-11}	-0.1094	0.845	0.6839	$-8.5, 1.1 \times 10^{-17}$
sentiment	LightGBM causal-tuned	0.13037	1.254×10^{-6}	4.611×10^{-11}	-0.0633	0.867	0.6912	$-8.0, 7.6 \times 10^{-16}$
market_ew	LightGBM causal-tuned	0.13038	1.250×10^{-6}	4.321×10^{-11}	0.0034	0.867	0.7121	$-6.9, 3.2 \times 10^{-12}$
market_ew	XGBoost causal-tuned	0.13043	1.250×10^{-6}	4.337×10^{-11}	-0.0002	0.864	0.7117	$-7.2, 3.0 \times 10^{-13}$
sentiment	XGBoost causal-tuned	0.13049	1.260×10^{-6}	4.813×10^{-11}	-0.1099	0.853	0.6809	$-8.4, 3.8 \times 10^{-17}$
implied_vol	XGBoost causal-tuned	0.13077	1.263×10^{-6}	4.636×10^{-11}	-0.0692	0.842	0.6978	$-5.7, 9.4 \times 10^{-9}$
moments	ridge causal-tuned	0.13084	1.249×10^{-6}	4.260×10^{-11}	0.0175	0.936	0.7029	$-19.2, 3.3 \times 10^{-82}$
moments	ElasticNet causal-tuned	0.13114	1.253×10^{-6}	4.201×10^{-11}	0.0311	0.933	0.7074	$-17.6, 5.2 \times 10^{-70}$
baseline	LightGBM causal-tuned	0.13118	1.259×10^{-6}	4.508×10^{-11}	-0.0396	0.870	0.6978	$-5.5, 2.2 \times 10^{-8}$
liquidity	ridge causal-tuned	0.13120	1.274×10^{-6}	4.345×10^{-11}	-0.0020	0.898	0.7029	$-13.3, 9.5 \times 10^{-41}$
baseline	XGBoost causal-tuned	0.13162	1.262×10^{-6}	4.601×10^{-11}	-0.0610	0.855	0.6956	$-5.4, 4.1 \times 10^{-8}$
moments	Lasso causal-tuned	0.13178	1.256×10^{-6}	4.224×10^{-11}	0.0258	0.938	0.7052	$-8.8, 6.5 \times 10^{-19}$
liquidity	ElasticNet causal-tuned	0.13184	1.276×10^{-6}	4.320×10^{-11}	0.0036	0.903	0.7036	$-11.4, 2.2 \times 10^{-30}$
liquidity	Lasso causal-tuned	0.13206	1.278×10^{-6}	4.336×10^{-11}	0.0001	0.901	0.7030	$-11.2, 3.9 \times 10^{-29}$
implied_vol	ridge causal-tuned	0.13252	1.262×10^{-6}	4.240×10^{-11}	0.0221	0.921	0.7063	$-11.2, 1.8 \times 10^{-29}$
implied_vol	ElasticNet causal-tuned	0.13262	1.264×10^{-6}	4.202×10^{-11}	0.0308	0.920	0.7091	$-10.6, 2.5 \times 10^{-26}$
implied_vol	Lasso causal-tuned	0.13274	1.263×10^{-6}	4.183×10^{-11}	0.0351	0.923	0.7100	$-9.8, 9.9 \times 10^{-23}$
market_vw	ridge causal-tuned	0.13290	1.267×10^{-6}	4.248×10^{-11}	0.0201	0.928	0.7048	$-9.1, 8.5 \times 10^{-20}$
market_vw	ElasticNet causal-tuned	0.13315	1.266×10^{-6}	4.226×10^{-11}	0.0253	0.934	0.7056	$-7.3, 2.3 \times 10^{-13}$
market_ew	ridge causal-tuned	0.13319	1.264×10^{-6}	4.221×10^{-11}	0.0265	0.936	0.7057	$-7.2, 5.1 \times 10^{-13}$
market_vw	Lasso causal-tuned	0.13321	1.267×10^{-6}	4.231×10^{-11}	0.0242	0.928	0.7060	$-6.4, 1.1 \times 10^{-10}$
vol_demand	ridge causal-tuned	0.13337	1.278×10^{-6}	4.295×10^{-11}	0.0094	0.925	0.7018	$-4.0, 4.3 \times 10^{-5}$
market_ew	ElasticNet causal-tuned	0.13344	1.266×10^{-6}	4.214×10^{-11}	0.0280	0.936	0.7062	$-5.4, 5.5 \times 10^{-8}$
sentiment	ridge causal-tuned	0.13347	1.276×10^{-6}	4.276×10^{-11}	0.0138	0.934	0.7021	$-5.6, 1.3 \times 10^{-8}$
market_ew	Lasso causal-tuned	0.13352	1.268×10^{-6}	4.220×10^{-11}	0.0266	0.934	0.7060	$-4.6, 4.0 \times 10^{-6}$
vol_demand	Lasso causal-tuned	0.13359	1.276×10^{-6}	4.261×10^{-11}	0.0173	0.930	0.7036	$-3.3, 8.1 \times 10^{-4}$
vol_demand	ElasticNet causal-tuned	0.13361	1.276×10^{-6}	4.268×10^{-11}	0.0155	0.932	0.7027	$-3.4, 6.0 \times 10^{-4}$
sentiment	ElasticNet causal-tuned	0.13377	1.276×10^{-6}	4.247×10^{-11}	0.0204	0.940	0.7033	$-3.5, 3.9 \times 10^{-4}$
sentiment	Lasso causal-tuned	0.13390	1.277×10^{-6}	4.244×10^{-11}	0.0212	0.939	0.7037	$-2.0, 3.6 \times 10^{-2}$
baseline	ridge causal-tuned	0.13445	1.282×10^{-6}	4.301×10^{-11}	0.0080	0.933	0.7003	$5.0, 4.7 \times 10^{-7}$
baseline	ElasticNet causal-tuned	0.13470	1.281×10^{-6}	4.266×10^{-11}	0.0162	0.937	0.7024	$7.5, 6.0 \times 10^{-14}$
baseline	Lasso causal-tuned	0.13474	1.281×10^{-6}	4.262×10^{-11}	0.0171	0.937	0.7027	$7.0, 2.4 \times 10^{-12}$
<i>OLS-on-HAR incumbent (baseline)</i>	none (OLS)	0.13415	1.283×10^{-6}	4.336×10^{-11}	—	0.929	0.6984	(baseline)